

HARSH SACHAN

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EDUCATION

Galgotias College of Engineering and Technology

2022 – 2026

B.Tech in Computer Science Engineering (AI Specialization) · CGPA: 8.0

TECHNICAL SKILLS

Programming: Python, C++, SQL, JavaScript

Frontend Development: HTML5, CSS3, React.js

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

Machine Learning: Regression, Classification, Clustering, Model Evaluation, Hyperparameter Tuning

Deep Learning: TensorFlow, Keras, CNN, RNN, Transfer Learning

Generative AI: LLMs, LangChain, RAG, Vector Databases

Data Analytics Tools: Power BI (DAX, Power Query, Power Pivot), Excel (Pivot Tables, Power Query, Dashboard)

Database: MySQL, PostgreSQL, Advanced SQL (Joins, Subqueries, Aggregations, Foreign Keys)

Tools: Git, GitHub, MLflow (Basics), Streamlit, psycpg2

PROJECTS

Smart Inventory & Billing System

Python, PostgreSQL, Streamlit, psycpg2, SQL

- ▶ Built a full-stack inventory and billing management system using Python and PostgreSQL, with a normalized schema (customers, products, sales, sale_items) and foreign key relationships for data integrity.
- ▶ Implemented complete CRUD operations for customer, product, and sales modules, with automatic total amount calculation and full sales transaction history.
- ▶ Developed SQL analytics for total revenue by date range, top-selling products, and per-customer sales reports, all surfaced through an interactive Streamlit UI.

HR Attrition Analytics – Excel Dashboard

Excel, Pivot Tables, Power Query, Star Schema, Dashboard

- ▶ Built an HR attrition analytics workbook using a star-schema model (Fact_Attrition, Dim_Employee, Dim_Date, Dim_Satisfaction, Dim_Performance) to track employee turnover by department, gender, overtime, and job satisfaction.
- ▶ Created 6 pivot tables analyzing attrition rate, average satisfaction, average performance by year, and department-wise overtime trends.
- ▶ Designed an interactive Excel dashboard consolidating all insights with slicers for dynamic filtering across HR dimensions.

AI & ML Jobs Market Analytics – Power BI Dashboard

Power BI, DAX, Power Query, Data Modeling

- ▶ Designed a 4-page interactive Power BI report (Overview, Location Insights, Company & Role Insights, Skill & Hiring Insights) to analyze global AI/ML job market trends.
- ▶ Built a star-schema data model with 6 tables and engineered DAX measures for Total Jobs, Avg Salary, Median Salary, Remote Job %, and Avg Salary by Experience Level.
- ▶ Implemented drill-through filtering and slicers across experience level, country, region, and company rating for executive-ready storytelling.

TravelPal – Ride Sharing App

React.js, Redux, Node.js, Express.js, MongoDB, Material UI, Sass

- ▶ Built a full-stack carpooling web application inspired by BlaBlaCar, enabling users to create and search rides, manage profiles, and connect with co-travellers.
- ▶ Architected a React + Redux frontend with Material UI and Sass, and a Node.js/Express REST API backend connected to MongoDB Atlas.
- ▶ Implemented secure user authentication using bcrypt, with endpoints for user registration, login, ride management, and car management.

CarValue Predictor

Python, Scikit-learn, Streamlit, Pandas, NumPy

- ▶ Developed an ML web application to predict used car resale prices using Decision Tree Regressor with GridSearchCV, achieving ~84% R² score.
- ▶ Built a modular Streamlit app with cached model loading and preprocessing pipelines for fuel type, seller type, and transmission.

Personality Deduction App

Python, Logistic Regression, Streamlit

- Built an ML-powered app predicting Extrovert, Introvert, or Ambivert personality types using Logistic Regression with StandardScaler, achieving 99.75% test accuracy.

AI Loan Approval System

Python, Naive Bayes, Streamlit

- Developed a loan approval prediction system using Gaussian Naive Bayes achieving 84.97% accuracy, with a responsive Streamlit interface for instant predictions.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Data Science Professional
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
- IBM SkillsBuild – Introduction to Artificial Intelligence
- HackerRank – SQL (Advanced), Problem Solving, Software Engineer Intern
- Google Cloud – Introduction to Generative AI

ACHIEVEMENTS

- Built and deployed multiple AI, ML, SQL, Power BI, and Excel analytics projects with real-world applications.
- Designed end-to-end dashboards in Power BI and Excel using star-schema modeling, DAX, and Pivot Tables.
- Developed a full-stack inventory system with PostgreSQL and a ride-sharing platform using the MERN stack.
- Strong problem-solving skills in Data Structures and Algorithms.
- Experience with cloud deployment via Streamlit Cloud and GitHub.